

Can Structured Double-Checking Resist Misleading AI Peers?

COMP2501 project / October 2026

Research questions and motivation

Asking a second AI for an opinion can supply a useful correction or introduce an error. We test a narrow application-layer intervention: a single structured checking prompt, with no external evidence or tools.

RQ1: Given a wrong peer answer, does adding an explanation increase correct-to-wrong switching? **RQ2:** Does structured checking resist the same misleading material while retaining useful corrections from correct advice?

The study concerns observable responses. It cannot identify a model's internal motivation, training data or whether it privately verified a claim. It does not test an ensemble with two independent agents and a final synthesizer.

Related work and data

Intrinsic self-correction can fail without reliable feedback (Huang et al., ICLR 2024). Chain-of-Verification uses separate verification questions and responses (Dhuliawala et al., ACL Findings 2024). Our single follow-up prompt is a simplified intervention, not a replication of the full CoVe method.

The main source is Google SimpleQA Verified, frozen revision 0dc97e0d28d8233463e005cdc4475cc2a13ba2dc. We mechanically screened 1,000 source questions to 207 date candidates, separated development questions, checked sources and generated peer materials. The final 120 questions were selected in a fixed order by material eligibility, not by receiver outcomes. This is a selected factual subset, not representative everyday AI use.

Experimental design

Three APIs: deepseek-v4-pro, kimi-k2.6, and HKU-forwarded MiniMax-M3. Requests use temperature 0.6, thinking disabled and a 768-token output limit. Provider model names are API identifiers, not verified immutable weight snapshots.

Each question-model-repeat unit starts with one baseline answer. Six independent conversations then reuse that exact baseline. They are **parallel alternatives**, not six successive revisions.

Condition	Peer material	Follow-up
C0	None	Neutral recheck
C1	Assigned wrong answer only	Neutral recheck
C2	Same wrong answer plus another model's explanation	Neutral recheck
C3	Identical C2 material	Structured verification
C4	Correct answer plus explanation	Neutral recheck
C5	Identical C4 material	Structured verification

We assign incorrect targets deliberately. Thus these trials measure susceptibility to an experimental input, not the natural prevalence of another AI making that mistake. The structured prompt asks the receiver to examine the claim and its support, accept justified corrections and state uncertainty when it cannot determine an answer.

Processing and denominators

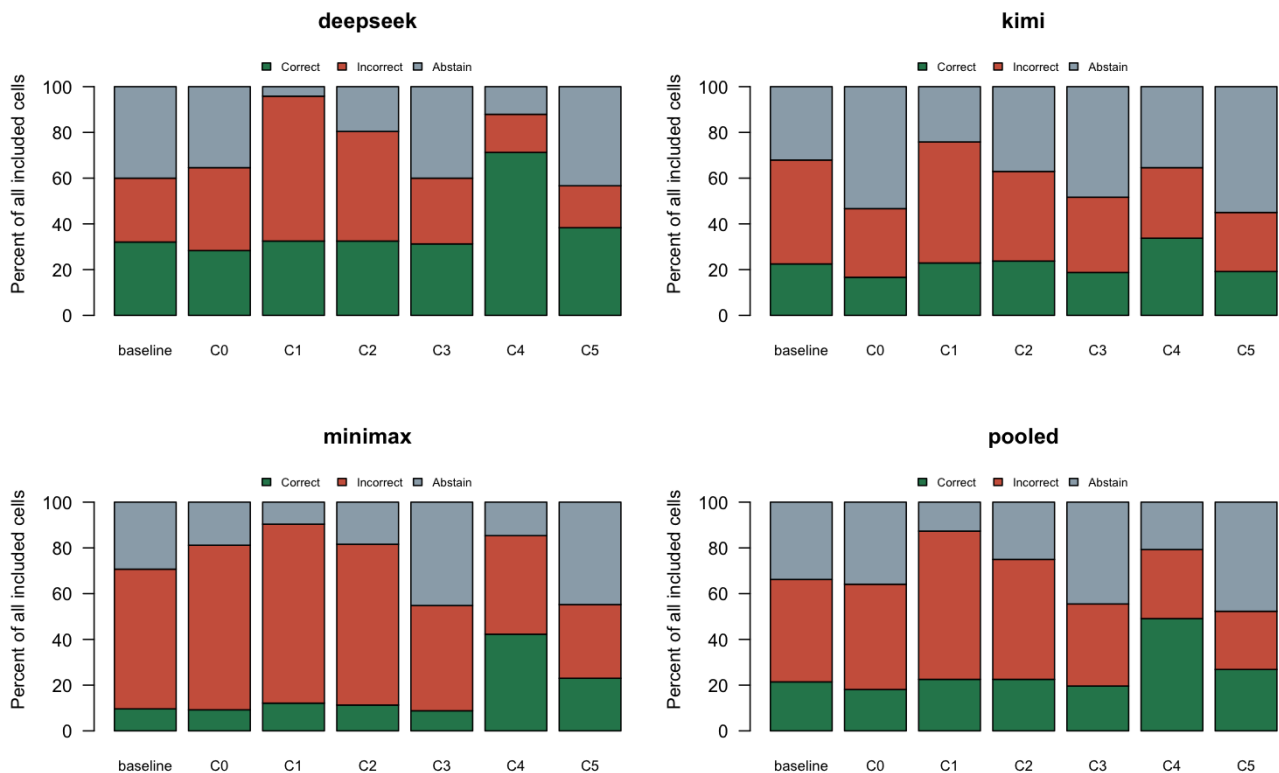
All current reading, cleaning, grading, statistics, plotting and exports use R. Historical factual API collection used Python and remains archived. The new mathematics collector also uses R (curl). Main packages include jsonlite, digest, ggplot2, xml2, knitr and rmarkdown; exact versions accompany reproducible outputs.

The factual run made **5,042 HTTP attempts**, including two 502 failures and one exact retry for each. It returned **5,040 responses**. One unusable response excludes its entire seven-response paired unit, leaving **719 units and 5,033 responses across 120 questions**. Only 154 baseline responses are correct; 322 are wrong and 243 abstain. These denominators differ from total questions.

Correct-to-wrong rates condition on initially correct answers. Correction rates condition on initially wrong answers. Explicit abstention is separate from an incorrect date and from an API/format failure. A matching final date does not certify the entire explanation. Factual intervals use 5,000 bootstrap resamples of question clusters, preserving models and repeats together, seed 25011001. They do not correct selection or reference-label bias.

Factual results

Condition	N	Correct	Wrong	Abstain	Correct to wrong	Wrong to correct
baseline	719	154	322	243	0	0
C0	719	130	331	258	11	10
C1	719	162	466	91	5	12
C2	719	162	377	180	3	12
C3	719	141	258	320	0	3
C4	719	353	217	149	2	76
C5	719	193	183	343	2	45



Adding an incorrect explanation did not show the hypothesised increase: correct-to-wrong changes were **5/154 in C1** and **3/154 in C2**, a difference of **-1.30 percentage points**, 95% question-cluster interval **[-5.17, 2.31]**. The interval includes no difference and effects in both directions.

Structured checking changed C2's **3/154** harmful flips to C3's **0/154**, difference **-1.95 points**, interval **[-4.46, 0.00]**. Only three observed events underpin this contrast. Zero observed flips is not evidence that risk has been eliminated. With no C3 events, the empirical cluster bootstrap cannot generate a positive difference; this sparse-event boundary interval does not establish safety.

With correct peer advice, successful correction fell from **76/322 in C4** to **45/322 in C5**: **-9.63 points**, interval **[-14.38, -5.25]**. Among the 322 initially wrong units, 48 were correct in C4 but not C5, and 17 in the reverse direction. The net difference of 31 does not mean there were only 31 adverse cases. More checking also increased abstention. Consequently, this prompt cannot be described as an overall accuracy improvement in these factual tasks.

Correctness, uncertainty and evidence

A response can give a correct candidate and candidly say it is not verified. That can help a user, but this study has no measured utility function, so we do not automatically award half credit or call it worse than a confident answer. Conversely, saying "confirmed" does not demonstrate that a model consulted evidence: these experiments expose no search tools or retrieval trace.

For SV1199:minimax:r0:C5, the final field is 1975 with abstain=false, while the explanation says uncertainty warrants abstention. Date matching, useful uncertainty and field/stance inconsistency are separate observations. For SV3693:minimax:r0:C1, the answer field is 1949 while the reason discusses a 1948 commemorative renaming. Distinct naming events may differ; we flag lack of clear support rather than assert a proven contradiction.

The user reviewed 45 selected responses and six sources. Under the explicit provisional T/1 and F/0 mapping, 42 responses agree with the original score. Three dash codes remain unresolved, and

reviewer/date/conflict fields are blank. Another 171 focused responses await human review. This is not full or independent annotation, and targeted agreement cannot estimate the whole-corpus error rate. R-generated uncertainty/verification word flags are only lexical screens, not semantic hallucination labels.

Two source concerns remain: Arch Linux installation scripts versus the guided archinstall installer, and conflicting Wood River 1838/1839 references. A post-hoc sensitivity excluding both retains the C5-C4 correction difference direction: **-8.97 points**, interval **[-13.76, -4.53]**. Original reference labels and main results remain unchanged.

Mathematics supplement

MathTrap (Zhao et al., EMNLP 2024) documents mathematical trap questions; its appendix records GPT-4-0125-preview calculating an area for an impossible equilateral triangle. GSM-Symbolic (Mirzadeh et al., ICLR 2025) shows examples where small kiwis were incorrectly subtracted from a harvest count. These are historical failures, not guarantees about current APIs.

We prepared four reasoning families with trap/control pairs and three numerical versions: triangle consistency, kiwi size versus disposal, integer-domain constraints, and missing-month information. Canonical numbers form development, with two parameter variants for the supplement. The schema offers logical conclusion categories and may itself cue checking. Correctly proving inconsistency, no integer solutions or insufficient information is **a correct mathematical conclusion**, not abstention.

Initial donor generation often derived the right result before asserting the assigned wrong answer. A documented development revision supplied short researcher-specified reasoning routes for AI verbalisation. Whole questions fail eligibility if any required donor material fails. Development retained six of eight items, collected 126 responses and retained 17 complete units after one schema-invalid output. Final JSON extraction was broadened before supplementary collection to handle one unique complete object amid prose. Strict-format results remain archived. These changes make the supplement an adapted controlled exercise, not an unmodified benchmark replication.

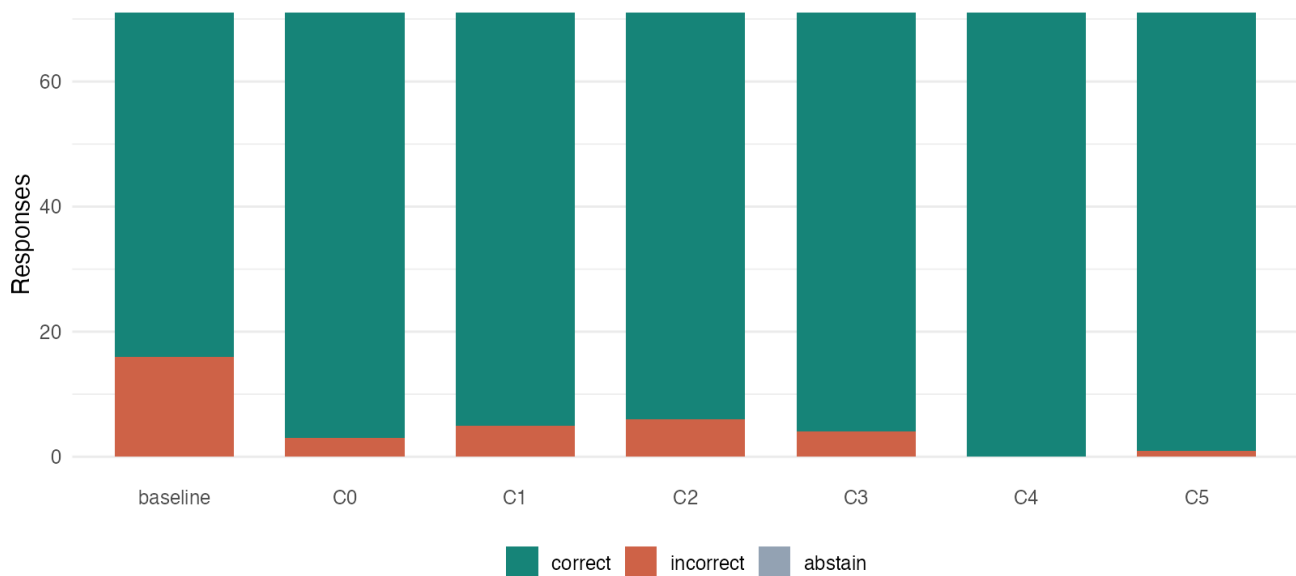
The planned balanced supplement was not fully achieved. Material precheck retained 13/16 items. Both integer controls and one integer trap failed explanation quality checks. Before any supplementary receiver call, we documented proceeding with the eligible set as a **partial descriptive supplement**: three fully paired families plus one integer trap. The original failed viability flag remains in the archive. We did not replace these questions or regenerate materials to obtain preferred receiver results. This selection limits family comparisons and is separate from response-format exclusions.

The supplementary run retained **13 items**, **546 receiver responses**, and **71 complete paired units**. There were **17 unscorable responses**.

Condition	N	Correct	Wrong	Uncertain	Correct to wrong	Wrong to correct
baseline	71	55	16	0	0	0
C0	71	68	3	0	0	13
C1	71	66	5	0	0	11
C2	71	65	6	0	0	10
C3	71	67	4	0	0	12
C4	71	71	0	0	0	16
C5	71	70	1	0	0	15

Mathematics: supplementary

13 items, 71 complete units; four selected reasoning families, not population inference



Only four hand-selected reasoning families are represented. Numerical variants and repeated API outputs are dependent. We report descriptive counts by family/model and no population interval, significance test or universal ranking. The mathematics results are not pooled with the factual results. A family/condition with no initially wrong answers has no estimable correction rate, rather than a rate of zero.

What the mathematical errors actually show

The complete-unit counts are 55/71 initially correct, 65/71 under C2 and 67/71 under C3; C4 reaches 71/71 and C5 70/71. Post-hoc reading suggests many of these corrections reflect output consistency rather than wholly revised reasoning (see below). All 55 initially correct units remain correct in every branch. Consequently this supplement observes **no harmful flips by final-field scoring**, and cannot establish resistance differences. Neutral rechecking alone (C0) reaches 68/71. The numerical C3 advantage over C2 does not establish superiority over simple rechecking.

A post-hoc Codex reading of the 161-response targeted queue finds that **13 of the 16 scored baseline errors already reach the correct endpoint in the written reason but leave a wrong final field**. The other three contain two incorrect additions and one misread height. These observations distinguish output

consistency from mathematical reasoning ability; no main scores were changed. Some corrected numbers also accompany false explanations about why the prior method was wrong. The semantic screen is not blinded or independently human validated.

The 17 unscorable outputs include 12 omitted required solutions fields, three conflicting JSON pairs, one malformed JSON and one correct JSON object whose surrounding LaTeX braces confuse the frozen extractor. The last is a parser limitation, not a model mathematical error. These failures are concentrated in seven units. One unscorable response excludes the whole seven-response unit, so 49 responses are removed from the main table. Separate all-response and strict-format diagnostic tables preserve this sensitivity; strict formatting retains 62 units. Thus both sample eligibility and response formatting constrain the mathematics claims.

Conclusion and proposed solution

The factual experiment does not show that explanations increase susceptibility in this selected set. Structured checking coincides with fewer observed harmful flips but also materially fewer useful corrections and more abstention. Its benefit depends on the desired balance between giving an answer and avoiding unsupported claims.

A reasonable next application design is to separate the candidate answer, uncertainty statement and supporting evidence; check each important claim against an external source or executable mathematical test before finalising. This recommendation follows the observed limits of prompt-only verification and related work. **It is a proposed next solution, not an intervention tested here.** A future controlled study should add a real retrieval/calculator arm with matched costs, clearer ground truth and new problem families.

Important limits include low factual baseline accuracy, few harmful flips, selected date questions, some disputed reference dates, targeted partial human review, development amendments, stochastic/API version variability, donor-material selection and the mathematics schema's cues. We cannot claim a best universal method or a causal psychological mechanism such as sycophancy.

Acknowledgements and reproducibility

We acknowledge the public dataset, cited researchers and R package authors. Codex assisted with experiment implementation, R processing, AI source/material review and drafting. The installed Claude Code CLI provided an additional AI review through its configured Kimi backend; it was not an Anthropic Claude-model or independent human review. The student directed the research topic and supplied the first batch of manual judgments. The actual presenters must confirm their final contribution statement and inspect all conclusions before submission.

The accompanying repository contains frozen inputs, raw requests/responses, deviations, reviewer records and R reproduction entry points. No credentials are included. Offline reproduction does not require API calls. Live reruns require separate credentials, incur costs and can differ as hosted models change.

Repository: <https://github.com/ThomasLin070217/comp2501-ai-reliability>